Fiscal Intelligence: Python-Driven Analysis of a $103M Budget

Data Source Disclosure: Modeled dataset. Figures are illustrative and are not the reported financials of the American Bar Association or any other organization.

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Executive Summary

This project presents a Python-based (Pandas/NumPy) fiscal audit framework applied to a modeled three-year dataset representing a $103M public-interest organization, with line items structured on American Bar Association (ABA) and State Bar financial statements. The figures are illustrative and are not reported financials. The analysis automates the algorithmic detection of fiscal red flags by evaluating liquidity trends, operating deficits, reserve floors, and revenue volatility. In the modeled data, operating deficits reach $28.8M in FY 2024, reserves fall to 11.8% of operating expenses against a 17% benchmark, and federal VOCA funding drops 45%, while liquidity declines but stays adequate at 263%.

Metric	FY 2022	FY 2023	FY 2024	Benchmark / Note
Total Revenue	$103.20M	$96.30M	$89.20M	-13.6% over 3 yrs
Operating Expenses	$101.70M	$116.30M	$118.00M	+$16.30M (+16.0%)

Metric	FY 2022	FY 2023	FY 2024	Benchmark / Note
Operating Deficit / Gap	+$1.50M	-$20.00M	-$28.80M	-$48.80M cumulative
Liquidity Ratio	321%	286%	263%	Adequate (>100%)
Reserve Capacity %	17.2%	11.8%	11.8%	GFOA benchmark: 17.0%
Federal VOCA Grants	$35.00M	$35.00M	$19.25M	45% Cut in FY 2024

Key Findings

- **Operating Deficits:** Annual operating gaps emerged in FY 2023 (-$20.0M) and widened in FY 2024 (-$28.8M), accumulating a $48.8M deficit over two years as spending rose 16.0% ($101.7M to $118.0M) while total revenue fell 13.6%.
- **Reserve Shortfall:** Working capital reserve capacity fell below the GFOA-recommended best practice benchmark of 17.0% (2 months of operating expenses), dropping from 17.2% in FY 2022 to 11.8% in FY 2023 and FY 2024. At the FY 2024 level, reserves are $6.1M short of the benchmark.
- **Grant Volatility Dependency:** Federal VOCA funding equaled 34% of total revenue in FY 2022. A 45% reduction (from $35.0M to $19.25M) produced a coefficient of variation of 0.25, above the 0.20 screening threshold, which indicates exposure to a single federal funding source.
- **Liquidity Dynamics:** The liquidity ratio decreased from 321% in FY 2022 to 286% in FY 2023 and 263% in FY 2024. While declining, a current ratio of 263% remains adequate for short-term obligation coverage.

Technical Methodology & Benchmarks

The automated audit engine evaluates fiscal health across three core parameters: (1) Reserve Floor Compliance, anchored to the Government Finance Officers Association (GFOA) recommendation of maintaining unrestricted budgetary reserves equal to at least 16.7%–17.0% of annual operating expenditures (~2 months); (2) Revenue Volatility, utilizing the Coefficient of Variation (CV = population standard deviation / mean) with an analyst-set screening threshold of CV > 0.20 to flag multi-year grant instability; and (3) Liquidity Coverage, monitoring the current ratio (current assets / current liabilities) against a 100% minimum.

Methodological Limitations

This analysis relies on a modeled dataset rather than observed records. The figures are illustrative, are scaled to a $103M public-interest organization, and follow the structure of American Bar Association (ABA) and State Bar financial statements. They are not the reported financials of the ABA or any other organization, and they are not municipal administrative records. The evaluation is constrained by a three-year historical baseline (FY 2022–2024), without predictive time-series forecasting or econometric modeling. With three observations, the coefficient of variation depends on the method: the population standard deviation gives 0.25 and the sample standard deviation gives 0.31, both above the 0.20 threshold. The GFOA reserve guidance is written for general-fund governments, so applying it to a bar association is an analogy. Alert thresholds (the 17% reserve benchmark and the 0.20 VOCA volatility coefficient) reflect analyst-set screening parameters rather than universal statutory limits.

Audit Script Engine (Python Code)

```python
import pandas as pd
import numpy as np

# 1. Modeled data (illustrative, scaled to a $103M organization; not actual ABA financials)
data = {
    'Fiscal_Year': [2022, 2023, 2024],
    'Total_Revenue': [103200000, 96300000, 89200000],
    'Operating_Expenses': [101700000, 116300000, 118000000],
```

```
    'Current_Assets': [32100000, 30000000, 26300000],
    'Current_Liabilities': [10000000, 10500000, 10000000],
    'Federal_Grants_VOCA': [35000000, 35000000, 19250000],
    'Reserves_Working_Capital': [17500000, 13700000, 13924000]
}
df = pd.DataFrame(data)

# 2. Alert parameters (set before running the analysis)
RESERVE_FLOOR_PCT = 17.0  # GFOA two-month benchmark (16.7%), rounded
CV_THRESHOLD = 0.20       # analyst-set screening threshold for grant volatility

# 3. Algorithmic fiscal metrics
df['Liquidity_Ratio_%'] = df['Current_Assets'] / df['Current_Liabilities'] * 100
df['Operating_Deficit'] = df['Total_Revenue'] - df['Operating_Expenses']
df['Reserve_Capacity_%'] = df['Reserves_Working_Capital'] / df['Operating_Expenses'] * 100
df['Reserve_Gap'] = (df['Reserves_Working_Capital']
                     - df['Operating_Expenses'] * RESERVE_FLOOR_PCT / 100)

# Volatility coefficient (CV = population std dev / mean, 3 observations)
voca = df['Federal_Grants_VOCA'].to_numpy()
volatility_coeff = np.std(voca) / np.mean(voca)

def usd(x):
    return f"-${abs(x):,.0f}" if x < 0 else f"${x:,.0f}"

# 4. Output findings
print("=== FISCAL RED FLAG AUTOMATED DETECTION ===")
gaps = df[df['Operating_Deficit'] < 0]
if not gaps.empty:
    items = [f"FY{str(y)[2:]} ({usd(d)})"
             for y, d in zip(gaps['Fiscal_Year'], gaps['Operating_Deficit'])]
    print("1. OPERATING DEFICIT: " + ", ".join(items))
last_fy = str(df['Fiscal_Year'].iloc[-1])[2:]
last_reserve = df['Reserve_Capacity_%'].iloc[-1]
```

```
if last_reserve < RESERVE_FLOOR_PCT:
    print(f"2. RESERVE SHORTFALL: FY{last_fy} reserve at {last_reserve:.1f}% "
          f"(benchmark: {RESERVE_FLOOR_PCT:.1f}%), {usd(df['Reserve_Gap'].iloc[-1])} vs. benchmark")
if volatility_coeff > CV_THRESHOLD:
    print(f"3. GRANT VOLATILITY: VOCA coefficient of variation = {volatility_coeff:.2f} "
          f"(threshold: {CV_THRESHOLD:.2f})")

trend = " | ".join(f"{y}: {r:.1f}%"
                   for y, r in zip(df['Fiscal_Year'], df['Liquidity_Ratio_%']))
latest_liq = df['Liquidity_Ratio_%'].iloc[-1]
status = "Adequate, above 100%" if latest_liq > 100 else "Below 100%"
print(f"\nLiquidity ratio trend: {trend} ({status})")
```

Audit Engine Console Output

```
None
=== FISCAL RED FLAG AUTOMATED DETECTION ===
1. OPERATING DEFICIT: FY23 (-$20,000,000), FY24 (-$28,800,000)
2. RESERVE SHORTFALL: FY24 reserve at 11.8% (benchmark: 17.0%), -$6,136,000 vs. benchmark
3. GRANT VOLATILITY: VOCA coefficient of variation = 0.25 (threshold: 0.20)

Liquidity ratio trend: 2022: 321.0% | 2023: 285.7% | 2024: 263.0% (Adequate, above 100%)
```

Technical Appendix

To view the live Python engine and raw data processing for this audit, click the link below to access the hosted development environment:

Click Here to Access the Live $103M Audit Notebook